parts of the world you should teach your kids to write computer code, while in others you had better teach them to draw fast and shoot straight.

Who owns the data?

If we want to prevent the concentration of all wealth and power in the hands of a small elite, the key is to regulate the ownership of data. In ancient times land was the most important asset in the world, politics was a struggle to control land, and if too much land became concentrated in too few hands – society split into aristocrats and commoners. In the modern era machines and factories became more important than land, and political struggles focused on controlling these vital means of production. If too many of the machines became concentrated in too few hands – society split into capitalists and proletarians. In the twenty-first century, however, data will eclipse both land and machinery as the most important asset, and politics will be a struggle to control the flow of data. If data becomes concentrated in too few hands – humankind will split into different species.

The race to obtain the data is already on, headed by data-giants such as Google, Facebook, Baidu and Tencent. So far, many of these giants seem to have adopted the business model of 'attention merchants'.[2] They capture our attention by providing us with free information, services and entertainment, and they then resell our attention to advertisers. Yet the data-giants probably aim far higher than any previous attention merchant. Their true business isn't to sell advertisements at all. Rather, by capturing our attention they manage to accumulate immense amounts of data about us, which is worth more than any advertising revenue. We aren't their customers – we are their product.

In the medium term, this data hoard opens a path to a radically different business model whose first victim will be the advertising industry itself. The new model is based on transferring authority from humans to algorithms, including the authority to choose and buy things. Once algorithms choose and buy things for us, the traditional advertising industry will go bust. Consider Google. Google wants to reach a point where we can ask it *anything*, and get the best answer in the world. What will happen once we can ask Google, 'Hi Google, based on everything you know about cars, and based on everything you know about me (including my needs, my habits, my views on global warming, and even my opinions about Middle Eastern politics) – what is the best car for me?' If Google can give us a good answer to that, and if we learn by experience to trust Google's wisdom instead of our own easily manipulated feelings, what could possibly be the use of car advertisements?[3]